

Windows OS systems, displaying file system integrity status of disk drives. It can fix logical file system errors and can also check the disk surface for physical errors or bad sectors. CHKDSK can be run from the Windows Shell, the Windows Command Prompt or the Windows Recovery Console.

Some general tips for recovering data: Don't delete files instantly. Move them to a temporary location such as the recycle bin before deciding whether you need them or not.

Mark important files as read-only. The OS will then warn you if you try to delete the file. Under systems that allow file system permissions, users can often only delete their own files. This prevents the erasure of critical system files or other's work.

1.7 Securely delete data

The reasons for deleting files are many: Freeing disk space, removing unnecessary or unneeded data, even making sensitive data unavailable to others. However, securely deleting data involves more than simply emptying the Recycle Bin. Basic file deletion methods only remove the direct pointers to data disk centres. A residual form of the data, called data remanence, still persists. Thus, anyone with data recovery tools can still unearth the data once more. Physical destruction may seem like the only way but thankfully, there exist many data erasure methods that remove information permanently while still keeping the disk operable.

Software-based overwriting is one such method. It writes patterns of meaningless data onto each of a hard drive's sectors. It differs from pure data erasure in that some data will still be intact and at risk of data breach or information theft. Nonetheless, data erasure employs multiple overwrites according to different overwriting standards. There are usually three types of data erasure that differ depending on the number of overwrites:

1. Fast erasure – Consists of one round of data deletion and the filling of space with random data.
2. Forced erasure – A US Department of Defense standard of file erasure. Unlike quick erase, data is overwritten with useless info 3 to 7 rounds in a row. Also referred to as the DoD 5220.22M Standard.